

Lab 2 - Introduction to Data

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1 Introduction

In this lab, we explore real-world baby naming data using the babynames dataset in R. This dataset contains information on baby names registered in the United States from 1880 onward, including the baby's name, gender, year of birth, and the number of babies given that name.

Our objectives are:

- To load and explore the dataset
- To analyze the popularity trends of a chosen name over time
- To identify top baby names by frequency
- To create rich visualizations comparing trends and gender-based preferences

2 Exercise 1 - Load and view dataset

We start by installing and loading the necessary packages: babynames, dplyr, and ggplot2

```
library(babynames)
library(dplyr)
library(ggplot2)
```

Now we take a quick look at the dataset:

```
head(babynames)
```

```
## # A tibble: 6 x 5
##   year sex  name      n  prop
##   <dbl> <chr> <chr>   <int> <dbl>
## 1  1880 F    Mary    7065 0.0724
## 2  1880 F    Anna    2604 0.0267
## 3  1880 F    Emma    2003 0.0205
## 4  1880 F  Elizabeth 1939 0.0199
## 5  1880 F   Minnie   1746 0.0179
## 6  1880 F  Margaret 1578 0.0162
```

```
str(babynames)
```

```
## tibble [1,924,665 x 5] (S3: tbl_df/tbl/data.frame)
## $ year: num [1:1924665] 1880 1880 1880 1880 1880 1880 1880 1880 1880 1880 ...
## $ sex : chr [1:1924665] "F" "F" "F" "F" ...
## $ name: chr [1:1924665] "Mary" "Anna" "Emma" "Elizabeth" ...
## $ n : int [1:1924665] 7065 2604 2003 1939 1746 1578 1472 1414 1320 1288 ...
## $ prop: num [1:1924665] 0.0724 0.0267 0.0205 0.0199 0.0179 ...
```

```
summary(babynames)
```

```
##      year      sex      name      n
## Min.   :1880   Length:1924665   Length:1924665   Min.    :  5.0
## 1st Qu.:1951   Class :character   Class :character   1st Qu.:  7.0
## Median :1985   Mode  :character   Mode  :character   Median : 12.0
## Mean   :1975                                     Mean  : 180.9
## 3rd Qu.:2003                                     3rd Qu.: 32.0
## Max.   :2017                                     Max.   :99686.0
##      prop
## Min.   :2.260e-06
## 1st Qu.:3.870e-06
## Median :7.300e-06
## Mean   :1.363e-04
## 3rd Qu.:2.288e-05
## Max.   :8.155e-02
```

```
range(babynames$year)
```

```
## [1] 1880 2017
```

```
length(unique(babynames$name))
```

```
## [1] 97310
```

3 Exercise 2 - Select a name

Let us choose the name “Emma” for our analysis. We will examine how its popularity has changed over the years.

```
emma <- babynames %>%
  filter(name == "Emma")
head(emma)
```

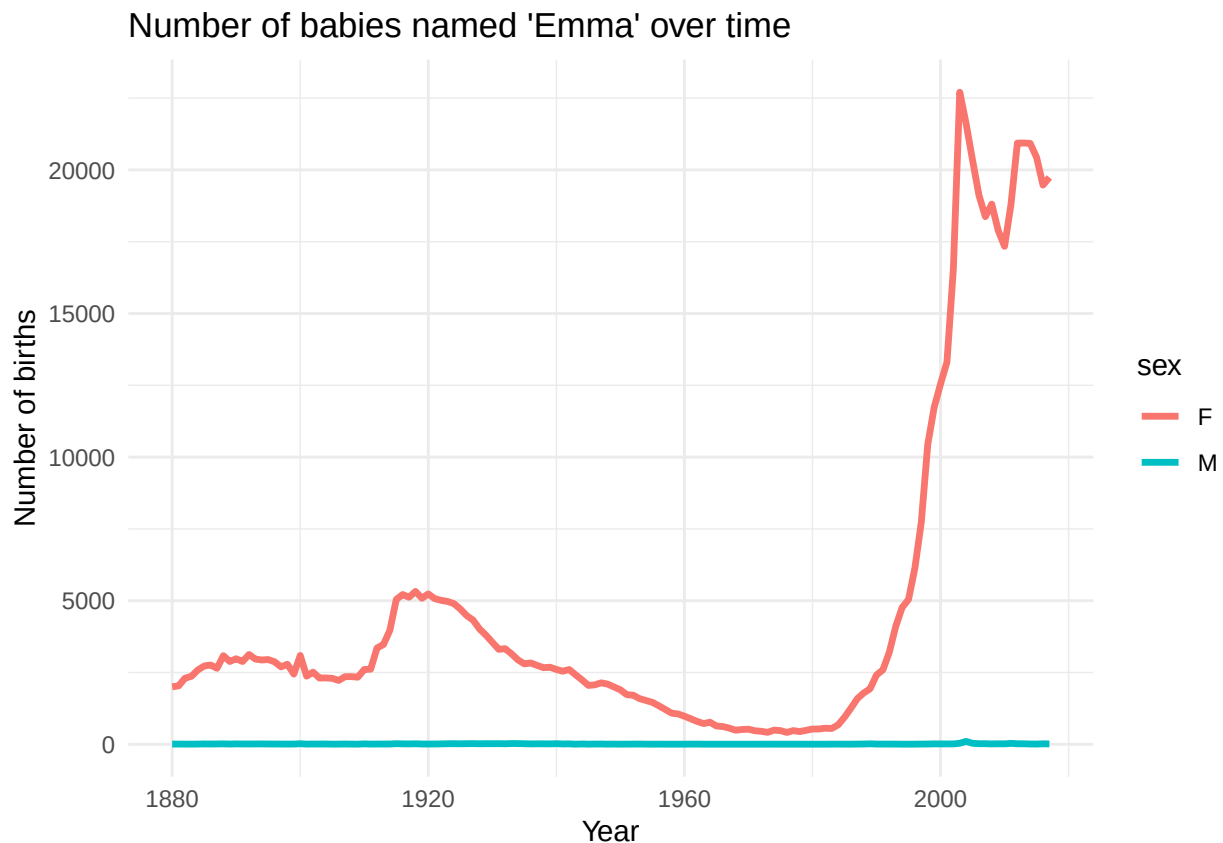
```
## # A tibble: 6 x 5
##   year sex  name      n      prop
##   <dbl> <chr> <chr> <int>   <dbl>
## 1  1880 F    Emma   2003  0.0205
```

```
## 2 1880 M Emma 10 0.0000845
## 3 1881 F Emma 2034 0.0206
## 4 1881 M Emma 9 0.0000831
## 5 1882 F Emma 2303 0.0199
## 6 1882 M Emma 7 0.0000574
```

4 Exercise 3 - Plot trend of the name

We now plot the number of babies named “Emma” over time, separated by gender.

```
ggplot(emma, aes(x = year, y = n, color = sex)) +
  geom_line(size = 1.2) +
  labs(
    title = "Number of babies named 'Emma' over time",
    x = "Year",
    y = "Number of births"
  ) +
  theme_minimal()
```



Observation: The name Emma saw a dramatic rise in popularity in the early 2000s, especially among girls.

5 Exercise 4 - Top 5 Names in Recent Years

We now determine the most popular names in the last decade (2010 to 2020), based on total number of babies given each name.

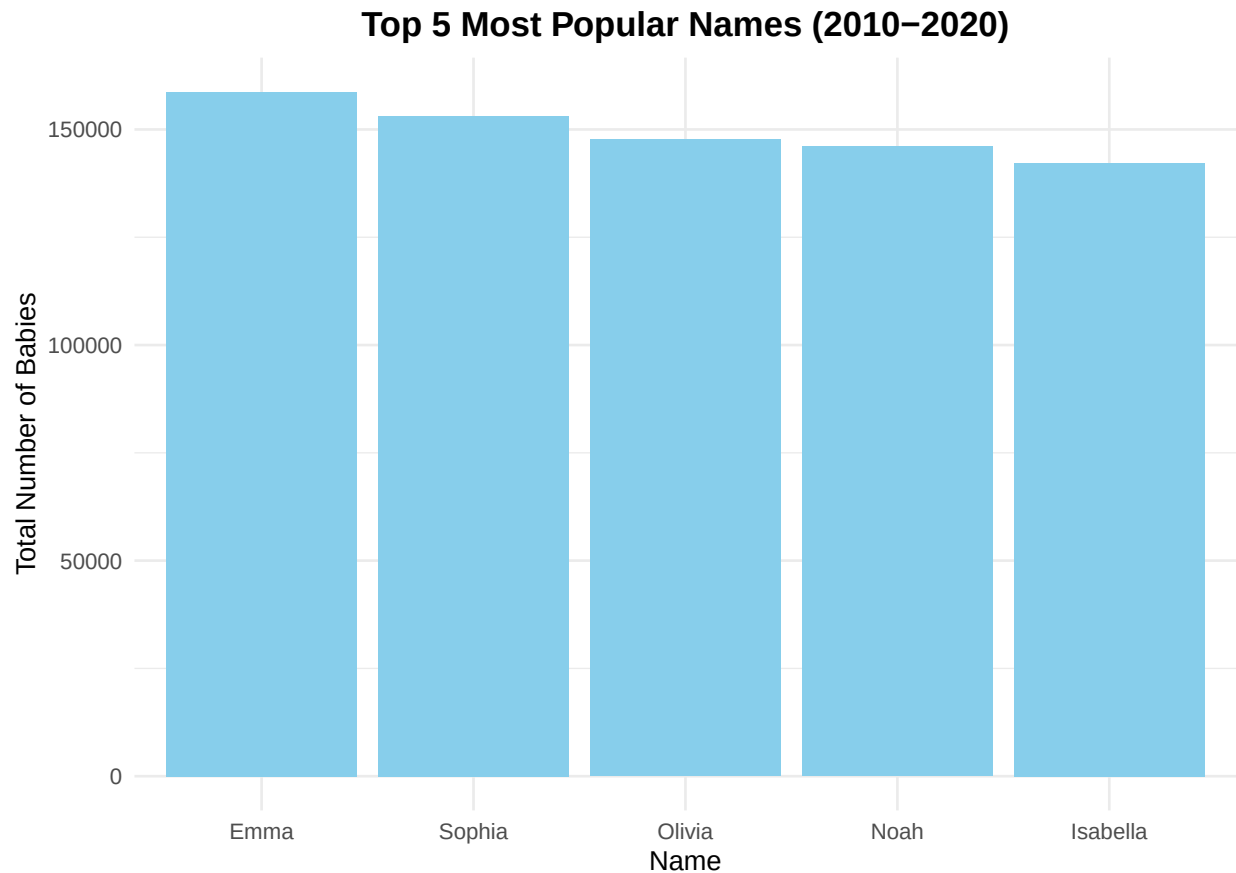
```
top_names_2010s <- babynames %>%  
  filter(year >= 2010 & year <= 2020) %>%  
  group_by(name) %>%  
  summarise(total_babies = sum(n)) %>%  
  arrange(desc(total_babies)) %>%  
  slice_head(n = 5)
```

```
top_names_2010s
```

```
## # A tibble: 5 x 2  
##   name      total_babies  
##   <chr>         <int>  
## 1 Emma          158715  
## 2 Sophia         153095  
## 3 Olivia         147630  
## 4 Noah           146059  
## 5 Isabella        142221
```

To visualize the Top 5 Names in the 2010s:

```
ggplot(top_names_2010s, aes(x = reorder(name, -total_babies), y = total_babies)) +  
  geom_bar(stat = "identity", fill = "skyblue") +  
  labs(  
    title = "Top 5 Most Popular Names (2010-2020)",  
    x = "Name",  
    y = "Total Number of Babies"  
  ) +  
  theme_minimal() +  
  theme(  
    plot.title = element_text(size = 14, face = "bold", hjust = 0.5)  
  )
```



6 Exercise 5 - Gender comparison for a specific year

We now look at the top 5 male and top 5 female names in 2017.

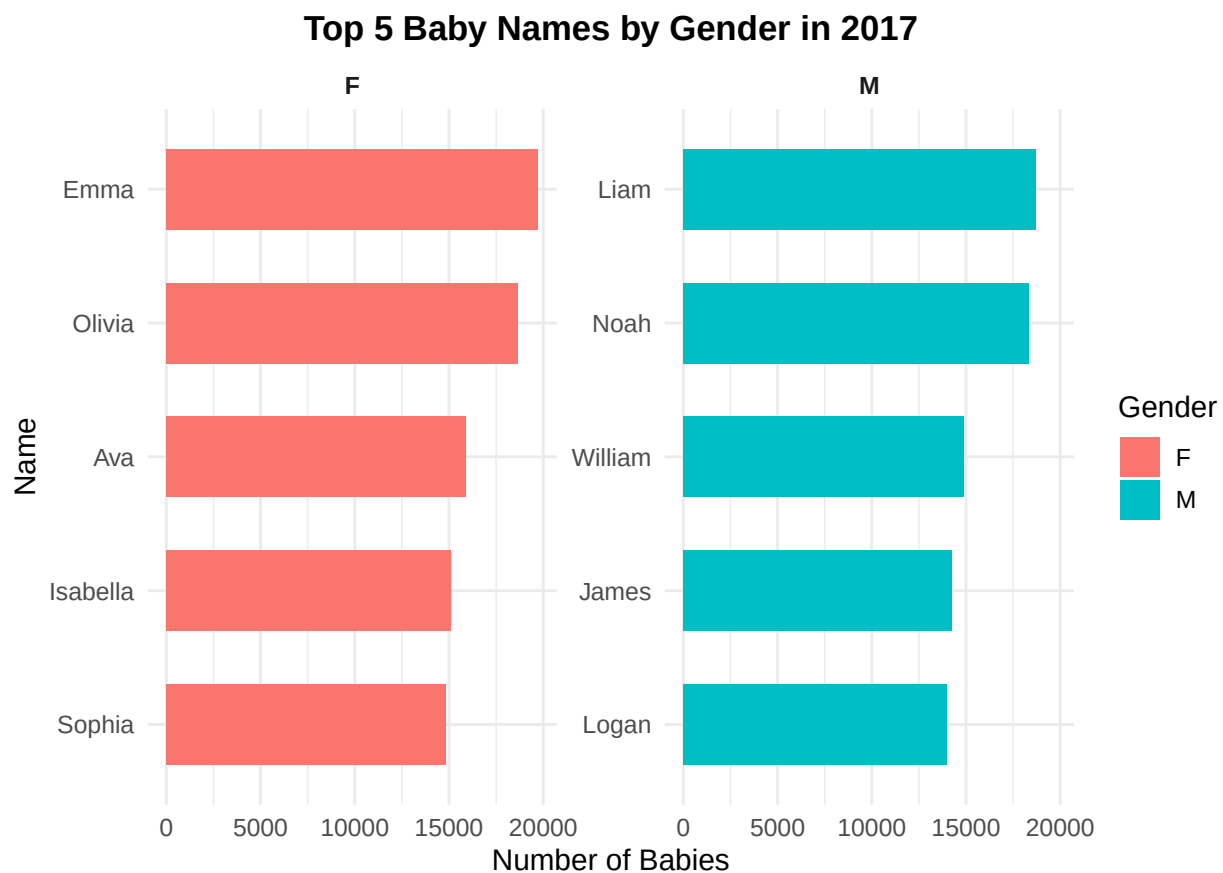
```
top2017 <- babynames %>%
  filter(year == 2017) %>%
  group_by(sex, name) %>%
  summarise(total = sum(n), .groups = "drop") %>%
  arrange(sex, desc(total)) %>%
  group_by(sex) %>%
  slice_head(n = 5) %>%
  ungroup()
top2017 # check the data
```

```
## # A tibble: 10 x 3
##   sex  name    total
##   <chr> <chr>    <int>
## 1 F    Emma    19738
## 2 F    Olivia  18632
## 3 F    Ava     15902
## 4 F    Isabella 15100
## 5 F    Sophia 14831
## 6 M    Liam    18728
```

```
## 7 M    Noah    18326
## 8 M    William 14904
## 9 M    James   14232
## 10 M   Logan   13974
```

To visualize Gender-Based Popular Names in 2017:

```
ggplot(top2017, aes(x = reorder(name, total), y = total, fill = sex)) +
  geom_col(show.legend = TRUE, width = 0.6) +
  facet_wrap(~ sex, scales = "free_y") +
  coord_flip() +
  labs(
    title = "Top 5 Baby Names by Gender in 2017",
    x = "Name",
    y = "Number of Babies",
    fill = "Gender"
  ) +
  theme_minimal(base_size = 12) +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    strip.text = element_text(face = "bold")
  )
)
```



Observation: In 2017, the top baby names for both genders reveal distinct naming preferences. For girls, Emma, Olivia, and Ava led the list, reflecting a continued popularity of short, soft-sounding names with vowel endings. For boys, names like Liam, Noah, and William dominated, which are traditional and timeless

choices. The difference in trends suggests that while female names tend to evolve with contemporary preferences, male names often maintain a more classic and stable pattern.

7 Conclusion

In this lab, we successfully:

- Explored a real-world dataset using `babynames`, Applied data wrangling techniques with `dplyr`, Used `ggplot2` for high-quality data visualizations
- Investigated long-term trends for a given name
- Identified top baby names by total count
- Compared gender-based naming trends for a given year

This foundational analysis sets the stage for more complex exploratory and predictive modeling in future labs.